Application: Loan repayment

- Consider a **fixed-rate** property loan.
- Suppose you are required to pay a specified stream of equal cash payments each month to a bank.
- You have taken out a 30-year loan with equal monthly payments.
- What would be your monthly payment if you take out a 30 year loan for 500,000 units at a "quoted" interest rate of 7.5% per annum?

monthly rate $= \frac{7.5}{12}\% = 0.625\%$

$$C_1 = \frac{C_0}{r}\left(1 - \frac{1}{(1+r)^t}\right)$$

$$\therefore 500000 = \frac{C_0}{0.00625}\left(1 - \frac{1}{1.00625^{360}}\right)$$

$$\therefore C_0 = \frac{500000 \times 0.00625 \times 1.00625^{360}}{1.00625^{360} - 1}$$

$$= 3496.07 \text{ per month.}$$

How to find 'principle' or 'interest' paid every month.

interest for first month $= 0.625\% \times 500000 = 3125$

$\therefore$ Principal paid $= 3496.07 - 3125 = 371.07$

$\therefore$ for second month, principal left $= 500000 - 371.07 = 499628.93$

now interest for second month $= 0.00625 \times 499628.93 =$

- In the first month, you paid $0.625\% \times 500,000 = 3125$ in interest.
- The principal repayment in the first month equals $= 3496.07 - 3125 = 371.07$.
- The principal remaining is thus $500,000 - 371.07 = 499,628.93$.
- The interest payment is on the remining principal.
- The next month, the interest payment $0.625\% \times 499,628.93 = 3122.68$ in interest.
- The principal repayment in the second month equals $= 3496.07 - 3122.68 = 373.39$.
- The principal remaining is thus $499,628.93 - 373.39 = 499,255.54$.

This helps while repaying the loan before the term expires.

- Perpetuities are imperfect approximations.
- Assumptions of fixed interest and growth rates and payments forever are assumptions that are rarely satisfied.
- Perpetuities are useful for providing quick back-of-the-envelope estimates.
- Can apply this to stock valuation.

- Consider a mature business with profits of 1 million next year.
- It's profit is expected to grow at a rate of 2% per annum.
- The firm faces a cost of capital of 8%.

What will be the present value.

$$\frac{1000000}{0.08 - 0.02} = \frac{1000000}{0.06} = 16.67 \text{ million}$$

for perpetuity $C_0 = \frac{C_1}{r - g}$

This is an estimate only.

Application: Stock Valuation

- The growing perpetuity formula is sometimes applied to the stock market.
- If one believes that (i) a stock's dividend will grow by $g = 5\%$ forever, (ii) the appropriate rate of return is $r = 10\%$, and (iii) the stock will pay dividends of D = 10 units next year, then one might feel that a stock price today of

$$\frac{10}{10\% - 5\%} = 200$$

would be reasonable.

- The growing perpetuity formula used this way is also known as the **Gordon growth model**.

Stock price $= \frac{\text{dividend}}{r - g} = \frac{10}{0.1 - 0.05}$

$= \frac{1000}{0.05} = 200$

☆ Gordon Growth Model.

Application: Stock Valuation

- The Gordon growth model has several applications.
- For instance, one can estimate the cost of capital for a company based on its dividend yield and its expected dividend growth rate.
- Consider a company which has forward-looking dividend yield of 2.2%.
- This **dividend yield** refers to the anaysts' consensus forecast of next year's dividend divided by the stock price, D/P.
- Thus for the company,

$$\frac{\text{Dividends D Next Year}}{\text{Stock Price P today}} = r - g = 2.2\%$$

- From the information about the forward-looking dividend yield, one can infer that the market believes that the appropriate cost of capital (r) for the company's equity exceeds its growth rate of dividends (g) by 2.2%.

$\frac{D}{r-g}$ = present price (P)

$\therefore \frac{D}{P} = r - g$.

$\therefore$ if forward-looking dividend rate = 2.2%

it means market believes $r - g = 2.2\%$.

- Analysts also looked at the summary of the company's cash flow statement and noted that the company paid 6.6 billion in dividends last year, which is almost the same as it paid out 4 years ago.
- From here, one estimated that 0% per year is an estimate of the future growth rate of the company's dividends.
- Thus the financial markets valued the company as if it had a per-annum cost of capital r of about 2.2%.
- Note that this is just an approximation, since growth rate and cost of capital are unlikely to remain constant forever.

Corporate Finance

Practice Problems

Problem 1A

Student A is trying to figure out the formula for the present value of a stream of payments of 1 unit a year, beginning 1 year from now, and continuing forever. He knows that the value is expressed by the infinite series

$$P = \frac{1}{1+r} + \frac{1}{(1+r)^2} + \dots$$

but cannot remember the formula for the sum of infinite geometric series. Student A however does know that if the first payment were to arrive now, rather than one year later, the present value of the sum would be 1 unit higher:

$$P + 1 = 1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} + \dots$$

Student B also does not remember the formula for the sum of geometric series. He is however aware that the present value of 1 unit a year forever, starting now has to be $(1+r)$ times as large as the present value of 1 unit a year, starting a year from now.

Can student A and B combine their knowledge to find out the value of P?

Student 'A'

$P + 1 = 1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} + \dots$

Student B. $P(1+r)$

$\therefore P(1+r) = 1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} + \dots$

$P(1+r) = P + 1$

$\therefore P + Pr = P + 1$

$\therefore r = \frac{1}{P}$ or $P = \frac{1}{r}$

Lets understand again. Lets say Rs. 100 is paid every month for 10 years at interest rate is 10%

What is the present value $= \frac{100}{1+.1} + \frac{100}{1.1^2} + \dots \frac{100}{1.1^{10}}$

$\therefore A = 100\left(\frac{1}{1.1} + \frac{1}{1.1^2} + \dots \frac{1}{1.1^{10}}\right)$

Lets say $X = \frac{1}{1.1} + \frac{1}{1.1^2} + \dots \frac{1}{1.1^{10}}$

$\therefore \frac{X}{1.1} = \frac{1}{1.1^2} + \dots \frac{1}{1.1^{11}}$

$\therefore X - \frac{X}{1.1} = \frac{1}{1.1} - \frac{1}{1.1^{11}}$ or $\therefore X = \frac{1.1^{10} - 1}{1.1^{11}} \times \frac{1.1}{1.1-1}$

$= \frac{1.1^{10} - 1}{1.1^{10}(1.1-1)}$

$\therefore A = \frac{100}{0.1}\left(\frac{1.1^{10}-1}{1.1^{10}}\right) = \frac{100}{0.1}\left(1-\frac{1}{1.1^{10}}\right) = \frac{C}{r}\left(1-\frac{1}{(1+r)^t}\right)$

Problem 2

In this problem, we will use the fact that all riskless assets must earn the same return to get an approximate answer of how to price a depletable resource such as oil, which we assume here to be a riskless asset. We assume that perfect market assumptions hold.

Problem 2A

Start by assuming that there is no cost to extract oil from the ground. We are are interested in characterizing how the price of oil will change over time.

1. Assuming that the interest rate is r, how should the price of oil in period t and $t+1$ be related? What does this imply about the growth rate of price of oil?
2. We are also interested in examining further what determines the price level of oil. Suppose that the demand for oil is constant at D barrels a year, and that there is a total supply of S barrels. How many years of oil are left? Denote this by T.
3. Once the oil has been depleted, one will have to use an alternative technology, say LG, which can be produced at a constant cost of C units per barrel and will be available from that point on. Assume that LG is a substitute for oil. T years from now, when the oil is just being exhausted, what should be the price of oil?
4. What should the price of oil today be as a function of C, T, r?
5. What happens to the price of oil today if there is an unforeseen new discovery of oil?
6. What happens to the price of oil today, if there is a technological breakthrough that decreases the value of C?

In no arbitrage condition.

$P_{t+1} = P_t + P_t \cdot r = (1+r) P_t$

$T = \frac{S}{D}$

$C = (1+r)^T P_0$

$\Rightarrow P_0 = \frac{C}{(1+r)^T}$

If $S \uparrow$, $T \uparrow$ & $\therefore P_0 \downarrow$

if $S \downarrow$, $T \downarrow$ & $P_0 \uparrow$

If $C \downarrow$, $P_0 \downarrow$

Problem 2B

Suppose now that there is a cost of X per barrel to extract oil from the ground. Let price in period t be denoted by p_t and let r be the interest rate.

1. If a firm extracts a barrel of oil in period t, how much profit would it make? → $P_t - X$
2. If a firm extracts a barrel of oil in period $t+1$, how much profit would it make? → $P_{t+1} - X$
3. What is the present value of the profits from extracting a barrel of oil in period $t+1$? → $\frac{(P_{t+1}-X)}{(1+r)^{t+1}}$
4. If a firm is willing to supply oil in each of the two periods, what must be the relation between the present value of profits of oil in period t and $t+1$? Solve for p_{t+1} as a function of p_t.
5. Is the percentage rate of price increase between periods t and $t+1$ the same as you found in part A?

$\Rightarrow \frac{P_t - X}{(1+r)^t} = \frac{P_{t+1} - X}{(1+r)^{t+1}}$

$\therefore (P_t - X)(1+r) = P_{t+1} - X$

$P_{t+1} = P_t(1+r) - Xr - X = P_t(1+r) - Xr$

$\therefore P_{t+1} = P_t(1+r) - Xr$

$\frac{P_{t+1} - P_t}{P_t} = \frac{P_t(1+r) - Xr - P_t}{P_t} = \frac{P_t r - Xr}{P_t} = r - \frac{Xr}{P_t}$

In first part $\frac{P_{t+1} - P_t}{P_t} = \frac{P_t + P_t r - P_t}{P_t} = r$

In second part it is smaller than this.

3 Present Value III

NPV

- Assume constant interest rates, no uncertainty and perfect markets.
- Recap of how to use NPV in deciding whether to take up a project.
- Translate cash flows at different points in time into the same units - money today - using the concept of present value.
- Take up projects which have a positive NPV.
- Reject projects which have a negative NPV.

- Why is NPV a good rule to use?
- In a perfect market with perfect information, a positive-NPV project is the equivalent of free money.
- Suppose you can borrow or lend money at 6% per year and you have an investment opportunity that costs 1 unit but yields 1.07 units next year, you can receive 0.01 units next year for free.
- If you can sell someone an investment opportunity for 1 units which yields only 1.05 units next year, you can get 0.01 units for free.
- Only zero-NPV projects (1 unit cost for 1.06 payoffs) do not allow one to earn free money.
- In a perfect world, if one has all the correct inputs, NPV is the appropriate decision benchmark.
- Information other than the NPV is redundant.

- In a perfect world with no uncertainty, positive NPV projects are rare.
- Otherwise one easily get rich.
- Everyone would want to take them up.
- Economy would adjust.
- Rate of return would be be bid up to the level where positive NPV projects are rare.

Why NPV?

- Despite its conceptual simplicity, NPV is hard to apply in the real world.
- One rarely knows cash flows and discount factors perfectly - one needs to estimate them.
- Also, perfect market assumptions are violated in the world.
- There are taxes, transactions costs, disagreements, buyers and sellers with market power.
- Deal with some of these complications later on in imperfect markets.
- Even in imperfects markets, NPV is an important benchmark.
- Other rules may provide additional useful information and may recommend alternative project choices.

- In a perfect capital market, when deciding on a project, should one's preference about timing of cash flows matter?
- Person A: wants money today, does not want to incur an upfront expensive which is involved in the project.
- Person B: want to defer consumption and save for the future.
- Would person A and person B agree on what to do with a project?

- Suppose one has 150 units cash on hand and one has exclusive access to a project that costs 100 units and returns 200 units next year.
- Appropriate interest rate is 10%.
- Would person A take up the project?

'A' invests 100 in project.

$\Rightarrow$ present value $= \frac{200}{1.1} = 181.8$

sells for 181.88

∴ he is left with $50 + 181.8 = 231.8$

- Person A should take up the project and incur cost of 100, and then sell it in the competitive market.
- Price one would get

$$\frac{200}{1.1} = 181.82$$

- Amount one gets today $181.82 - 100 + 150 \approx 231.82$.
- This is more than what one would have got if one had consumed 150 cash at hand.

- Suppose one has 150 units cash on hand and one has exclusive access to a project that costs 100 units and returns 200 units next year.
- Appropriate interest rate is 10%.
- Person B: wants to defer consumption and save for the future.
- Would person B take up the project?

'B' invests 100 in the project

Next year, he gets.

$50 \times 1.1 + 200 = 255$.

If he had not taken the project

then $150 \times 1.1 = 165$

∴ He should take up the project.

Imp. takeaway.

- Lesson: Regardless of when one wants cash (consumption decision), one is better off taking all positive-NPV projects (investment decision) and using the capital market to shift consumption to when one wants it.
- Consumption decisions should not influence investment decisions: **separation of decisions**.
- Separation of decisions does not always hold in imperfect markets, where one faces different borrowing and lending interest rates.

Sensitivity Analysis.

- In the real world, it is often impossible to get the NPV inputs perfectly correct.
- Is it worse to commit an error in estimating cash flows or in estimating the cost of capital?
- Do a **scenario analysis**: consider a simple project to learn how changes in estimates matter to ultimate present value.
- Scenario analysis is useful for managers who need to learn sensitive their estimated value is to reasonable alternative possible outcomes.
- This method is also known as **sensitivity analysis**.

- Example 1 (Short-term project): Assume that project will pay off 200 next year and the correct interest rate is 8%.
- Correct present value is

$$\frac{200}{1+8\%} \approx 185.19.$$

- Suppose one makes a 10% error in cash flow, so that one assumes it is 220 instead of the actual 200.
- Computed present value:

$$\frac{220}{1+8\%} \approx 203.70.$$

- Difference between 203.70 and 185.19 is 10% of 185.19.

$NPV_1 = \frac{200}{1.08} = 185.19$

If error of 10% error in cash flow (assume 220 instead of 200) then $NPV_2 = \frac{220}{1.08} \approx 203.70$.

$NPV_2 = 185.19 + 10\%$ of 185.

∴ error in cash flow leads to 10% ↑ in NPV

Less say 10% error in interest rate 8.8%

∴ $NPV_3 = \frac{200}{1.088} \approx 183.22$

Take away In shorter run → 1. Error in interest rate does not impact the NPV so much.

2. Error in cash flow estimation impacts the NPV.

Long Run

- Example 2 (Long-term project): Assume that project will pay off 200 in 30 years and the correct interest rate is 8% per year.
- Correct present value is

$$\frac{200}{(1+8\%)^{30}} \approx 19.88.$$

- Suppose one makes a 10% error in cash flow, so that one assumes it is 220 instead of the actual 200.
- Computed present value:

$$\frac{220}{(1+8\%)^{30}} \approx 21.86.$$

$NPV_1 = \frac{200}{(1.08)^{30}} \approx 19.88$

$NPV_2 = \frac{220}{(1.08)^{30}} \approx 21.86$

$NPV_3 = \frac{220}{(1.088)^{30}} \approx 15.93$

- **Lesson**: Both cash-flow errors and cost of capital errors are important for long-term projects. ★
- More difficult to estimate far-away future cash flows than to estimate discount rate demanded by investors today for for-away cash flows. ✓

We have seen 'NPV' till now

Internal Rate of Return

- Internal Rate of Return (IRR) is another common capital-budgeting method.
- It often leads to the same recommendation as NPV criterion.
- IRR compares the project's rate of return to the prevailing rate of return.
- Assume there is a project with cash flows that translate into a rate of return of 15% (investment of 100 today leading to a payoff of 115 next period).
- Suppose that the prevailing discount rate of 10%.
- Since 15% > 10%, intuitively the project should be taken up.
- NPV criterion would also recommend the same.

- How to compute the rate of return on a project that has different payments?
- Suppose an investment costs 100,000 and pays off 5,000 in one year, 10,000 in two years, and 120,000 in three years.
- What is the rate of return of this project?
- Need to extend the concept of rate of return where many inflows and outflows occur.
- Want to have this new 'rate of return' give the same number as the simple rate of return if there is only one inflow and one outflow.
- **Internal rate of return** (IRR) is such a measure.
- In the context of bonds, the internal rate of return is also known as **yield-to-maturity** (YTM).

$$0 = -100000 + \frac{5000}{1+IRR} + \frac{10000}{(1+IRR)^2} + \frac{120000}{(1+IRR)^3}$$

$$\therefore \frac{5000}{x} + \frac{10000}{x^2} + \frac{120000}{x^3} = 100000$$

Using Excel IRR $x = 11.14\%$

- If there is only one inflow and outflow, the IRR is the simple rate of return.
- Consider a project that costs 100 units today and pays 120 units next year.

$$-100 + \frac{120}{1+r} = 0$$

$$100r = 120 - 100 = 20.$$

$$\therefore r = \frac{20}{100.} = 0.2. \quad 20\%.$$

Internal Rate of Return

- When projects have many positive and negative cash flows, they can have multiple internal rates of return.
- Consider a project that costs 100,000 today, pays 205,000 one year from now and has environmental costs of 102,000 two years from now.
- This project has two internal rates of return, -15% and 20%:

$$-100,000 + \frac{205,000}{1+(-15\%)} + \frac{-102,000}{[1+(-15\%)]^2} = 0$$

and

$$-100,000 + \frac{205,000}{1+(20\%)} + \frac{-102,000}{[1+(20\%)]^2} = 0$$

- In some cases, there can be more than two IRRs.
- Note: computer spreadsheets may not give you more than one IRR even if multiple IRRs exist.

$$\therefore \frac{205000}{1+IRR} + \frac{-102000}{(1+IRR)^2} = 100000$$

$$\therefore 205000x^2 - 102000 - 100000x = 0$$

$$\Rightarrow 205x - 100x - 102 = 0$$

Here we will get 2 values for $x \Rightarrow (1+IRR)$

$\therefore$ 2 values of IRR. (in this case)

There can be more also.

- There can exist projects which have no IRR.
- Consider a project that yields 100 today and 200 in the next period.
- Such a project has no internal rate of return, since NPV formula is never zero.

$100 + \frac{200}{1+r} = 0$

$\therefore \frac{200}{1+r} = -100$

$\frac{200}{1+r}$ will always be positive.

∴ for this we can't find IRR.

(when there is no cash outflow, IRR can't be found)

- Consider a project that has a cost of 10,000 today, then pays 27,000 in one year, and finally requires a cleanup cost of 20,000 in two years.
- There's no rate of return at which its NPV would turn positive.
- So it has no IRR.
- Note: computer spreadsheets usually give an error if no IRRs exist.

$-10000 + \frac{27000}{1+r} - \frac{20000}{1+r^2} = 0$

$\Rightarrow -10x^2 + 27x - 20 = 0 \Rightarrow 10x^2 - 27x + 20 = 0.$

$\sqrt{b^2-4ac} = \sqrt{27^2 - 800} =$

for this 'r' can't be found.

- The **IRR capital-budgeting rule** states that if and only if an investment project's IRR is above the appropriate discount rate (the cost of capital quoted like a required interest rate) for the project, then the project should be taken.
- Cost of capital in this context is called the hurdle rate.
- In many common cases, IRR capital-budgeting rule gives the same answer as the NPV capital-budgeting rule.
- If there are no or multiple IRRs, it may not give the same answer as the NPV capital-budgeting rule.

- Suppose an investment costs 100,000 and pays off 5,000 in one year, 10,000 in two years, and 120,000 in three years.
- We had found that the IRR was 11.14%.
- The IRR capital budgeting rule states that if the prevailing cost of capital in the economy is 11.25%, one should not take this project.
- If prevailing cost of capital is 11%, one should take up the project.
- NPV also gives the same recommendation.

- If cash flows are opposite - an upfront positive cash flow, followed by several negative flow - this would be an example of investment financing.
- Consider an example where one receives 100,000 upfront and then pays out 5,000, 10,000 and 120,000 in successive years.
- One would now want to take this financing alternative if and only if the prevailing interest rate is above 11.14%

$100000 - \frac{5000}{1+r} - \frac{10000}{(1+r)^2} - \frac{120000}{(1+r)^3}$

we use for $r = 11.14\%$.

In this case, if interest rate > 11.14% go for this.

- What are the advantages of using IRR?
- IRR is a characteristic of a project's cash flows. ✓
1. IRR can be computed without having to look at financial markets, interest rates or costs of capital. ✓
2. IRR can also help in judging project profitability and allow one to asses the performance of a manager. ✓
3. Comparing the IRR to the cost of capital can help to understand how much buffer one has in terms of getting one's cash flow estimates wrong by a certain percentage without the error affecting the investment decision. ✓

- IRR works well if there is only one unique IRR. ✓
- It is safe to use when there is only one positive or only one negative cash flow. ✓
- Need to make sure to apply the capital budgeting rule correctly. ✓
- If the project requires an upfront outflow followed by inflows, one should take up the project if its IRR is above the cost of capital. ✓
- If the project requires an upfront inflow followed by outflows, one should take up the project if its IRR is below the cost of capital. ✓
- Difficult to implement this with multiple IRRs or no IRRs - need to rely on graph of NPV function ✓
- Also difficult to implement when there are multiple costs of capital (for instance short-term interest rate vs long term interest rate)

- The IRR criterion might be misleading when projects are mutually exclusive.
- Should one always prefer a project with 100% IRR to a project with a 10% IRR?

- Consider the following example with a rate of return equal to 5% per annum:

1. Project 1 is an investment opportunity of 5 and returning 10.
2. Project 2 is an investment opportunity of 1,000 and returning 1,100.

- Project 1 has an IRR of 100%, while Project 2 has an IRR of 10%.
- Project 1 has a NPV of 4.52, while project 2 has a NPV of 47.62.
- If one can only take up one project, one should take up project 2.

project 1 $-5 + \frac{10}{1+r} = 0.$

$\therefore -5 - 5r + 10 = 0.$

$\therefore r = \frac{5}{5} = 1$

$\therefore r = 100\%.$

project 2 $-1000 + \frac{1100}{1+r} = 0.$

$r = \frac{100}{1000} = 10\%.$

$NPV_1 = -5 + \frac{10}{1+0.05} = -5 + \frac{10}{1.05} = 4.52$

$NPV_2 = -1000 + \frac{1100}{1.05} = 47.61$

$\therefore$ IRR of project 1 >> IRR of project 2.

but NPV of Project 2 >> NPV of Project 1.

$\therefore$ IRR is used when NPVs are similar.

Profitability Index

- Another measure sometimes used in capital budgeting is the profitability index.
- **Profitability index** divides the present value of future cash flows by the project cost (the negative of the first cash flow).
- Project 1: Consider a project which has cash flow of -100 units today, 70 units 1 year from now, 60 units 2 years from now and 50 units 3 years from now.
- If the interest rate in the economy is 20 percent, one can compute the present value of future cash flows as approximately 128.94 units.

$-100 + \frac{70}{1.2} + \frac{60}{1.2^2} + \frac{50}{1.2^3} = 28.94$

$PI = \frac{\text{PV of future cash flows}}{-(\text{first cash flow})}$

PV of future cash flows : $\frac{70}{1.2} + \frac{60}{1.2^2} + \frac{50}{1.2^3} = 128.94$

$\therefore PV = \frac{128.94}{-(-100)} = 1.29$

- Profitability index equals

$$\frac{128.94}{-(-100)} \approx 1.29.$$

- Note that the profitability index is meaningful only if the first cash flow is a cash outflow.
- NPV of the project equals approximately 28.94 units.

- A positive NPV project usually has a profitability index above 1.
- if the first cash flow is a cash outflow, one can use either the NPV or the profitability index.
- Statements "$NPV > 0$" and "profitability index > 1" become the same.
- So the profitability index can give the correct answer in the situation involving one negative cash flow followed by all positive cash flows.

- Project 2: Consider a project which has cash flow of -10 units today, 21.14 units 1 year from now, 18.12 units 2 years from now and 15.10 units 3 years from now.
- If the interest rate in the economy is 20 percent, one can compute the present value of future cash flows as approximately 38.94 units.
- Profitability index equals

$$\frac{38.94}{-(-10)} \approx 3.89.$$

- Project 2's profitability index is higher because it requires less capital upfront.
- Intuitively, second project might be preferred to first project (even though both have the same NPV), because it requires less capital.

PV (cash flow) : $\frac{21.14}{1.2} + \frac{18.12}{1.2^2} + \frac{15.10}{1.2^3} = 38.94$

$\therefore PI = \frac{38.94}{10} = 3.89$

NPV of this project is also 28.94

but PI is higher. $\therefore$ this project may be preferred.

Take away : for same 'NPV', go for higher PI

- Project 3: Consider a project which has cash flow of -100 units today, 140 units 1 year from now, 120 units 2 years from now and 100 units 3 years from now.
- If the interest rate in the economy is 20 percent, one can compute the present value of future cash flows as approximately 257.87 units.
- Profitability index equals

$$\frac{257.87}{-(-100)} \approx 2.58.$$

PV (cash in flows) : 257.87

PI : $\frac{257.87}{100}$: 2.58.

- Profitability index of project 3 is still lower than that of project 2.
- However, NPV of project 3 (equal to 157.87) is higher than that of project 2.
- Profitability index is biased towards project which require lower upfront costs.
- Profitability index should be used in choosing among projects only if NPVs of the two projects are equal (or very similar).

main takeaway.

5. Capital Budgeting Rules II

Payback Rule

- Another capital budgeting rule that is often used is the payback rule.
- Under the payback rule, projects are better if one can recover one's original investment faster.
- Project 1: Requires an upfront cost of 5 now, and pays out 8 one year from now: payback period = 1 year.
- Project 2: Requires an upfront cost of 5 now, and pays out 4 one year from now, and 10 units two years from now: payback period = 2 years.
- Project 3: Requires an upfront cost of 5 now, and pays out 4 one year from now, 0 two years from now and 1,000,000 units three years from now: payback period = 3 years.
- Assume that the interest rate is 5 percent.

7:54 / 12:35

$NPV_1 = -5 + \frac{8}{1.05} = 2.61$

$NPV_2 = -5 + \frac{4}{1.05} + \frac{10}{1.05^2} = 7.88.$

$NPV_3 = -5 + \frac{4}{1.05} + 0 + \frac{1000000}{1.05^3}$

$= 863836.$

- Using the payback rule, Project 1 has the shortest payback period and is the best from the payback rule perspective; it has the lowest NPV among the three.
- Project 2 has the second smallest payback period and is the second-best based on the payback rule; it has the second lowest NPV among the three.
- Project 3 has the largest payback period and is the worst based on the payback rule; it has the highest NPV among the three.

main takeaway.

Payback Rule

- Payback rule is intuitively easier to explain - how many years will it take to get one's money back?
- Payback rule's emphasis on earlier cash flows helps firms set criteria when managers may not be trusted enough.
- Payback can help in case of an entrepreneur with limited capital, facing an imperfect capital market: in this case, cost of capital is quite likely to be high, and it is important to get back money in a short amount of time.
- In some cases, payback rule is likely to give same answer as using more sophisticated method such as NPV.
- Summary: payback period is more useful as a supplementary information, rather than main criteria to decide on projects.

6 Time Varying Rates of Return

- So far we have been assuming that the rates of return per unit of time is same regardless of what the investment time horizon is.
- 30-year bond offered the same rate of return per annum, as did a 1-year bond.
- This is not the case in real world.
- In early September 2024, Indian government 1-year bond offered a rate close to 6.8 percent, while a 30-year bond offered a rate close to 7.1 percent.
- At the same time, in the United States, 1-year bond offered a rate close to 4.4 percent, while a 30-year bond offered a rate close to 4.0 percent.
- In 2021, the difference was even larger: 1-year bond offered a rate close to 0.4 percent, while a 30-year bond offered a rate close to 1.9 percent.

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- Banks also offer different rates for deposits for different durations.
- Companies also need to know how to compare what they pay when their investors are willing to given them money for a project that promises returns in 1 years vs returns in 30 years.
- What is the role of inflation in determining these differences?

- Rate of return differ depending on when payments are made.
- Interest rate next year can be higher or lower compared to this year.
- Long term bonds also offer different interest rates than short term bonds.
- We can use the tools we studied previously to think about how to deal with these cases.

- **Example 1**: What is the two-year holding rate of return if the rate of interest is 30% in the first year and 20% in the second year?
- One can use the same compounding formula as before:

$$1 + r_{0,2} = (1 + 30\%)(1 + 20\%) = 1 + 56\%.$$

- The two-year holding rate of return is thus 56%.
- More generally

$$1 + r_{0,2} = (1 + r_{0,1})(1 + r_{1,2})$$

$P = 100$, $I_1 = 0.3$ ∴ After 1 year 130

$I_2 = 0.2$ ∴ after 2 year $130 + 26 = 156$

∴ 2 yr holding rate = 56%.

formula

$$1 + r_2 = (1 + r_1)(1 + r_2)$$

$$1 + r_2 = (1 + .3)(1 + .2)$$

$$1 + r_2 = 1.3 \times 1.2 = 1.56$$

$$\therefore r_2 = 0.56.$$

6 Time Varying Rates of Return

- **Example 2**: Suppose the 1 year return is 30% from year 1 to 2, 40% from year 2 to year 3, and 50% from year 3 to year 4.
- What is the holding rate of return for an investment beginning next year for 3 years?

$$1 + r_{1,4} = (1 + r_{1,2})(1 + r_{2,3})(1 + r_{3,4})$$

- Plugging in values, we get $r_{1,4} = 173\%$.
- Three year holding rate of return for an investment that takes money next year and returns money in 4 years is 173%.
- 100 units invested 1 year from now would become 273 units 4 years from now.

$$1 + r_4 = (1 + r_1)(1 + r_2)(1 + r_3)$$

$$1 + r_4 = 1.3 \times 1.4 \times 1.5$$

$$= 1.82 \times 1.5$$

$$1 + r_4 = 2.730$$

$$\therefore r_4 = 1.73. = 173\%.$$

∴ 100 Rs will become 273.

- We need a way to compare rates of return over different time horizons.
- Is a rate of return of 10% over 3 years better than a rate of return of 20% over 6 years?
- Most returns are quoted in terms of annualized rates.
- Average annualized rates of return is a unit of measurement for the rate at which money accumulates.

- If one were earning a total three-year holding rate of return of 173% over the three year period, what would be the annualized rate of return?
- Answer is not the average rate of return, given by $173\%/3 \approx 57.7\%$.
- A return of 57.7% for three years would correspond to a return of $1.577^3 - 1 \approx 292\%$.
- The answer of 57.7% ignored the compounded interest on the interest that one would earn after the first and second years.

$= 1 + 1.73 = (1+r)^3$

$\therefore \quad 1+r = \sqrt[3]{2.73}$

$r = 1.39 - 1 = 0.39 = 39\%$

- **Example 3**: What annualized rate of return should one expect from a 100 unit investment today that promises a return of 240 units in 30 years?
- First step is to compute the total holding rate of return.
- The total 30 year holding rate of return is given by

$$r_{0,30} = \frac{240 - 100}{100} = 140\%.$$

- The annualized rate of return is the rate r_{30} which, if compounded for 30 years, offers a 140% rate of return:

$$(1 + r_{30})^{30} = 1 + 140\%$$

which gives us $r_{30} \approx 2.96\%$.

Total return $= \frac{240-100}{100} = 140\%$.

$\therefore \quad 1 + 1.4 = (1+r)^{30}$

$\therefore \quad 1+r = (2.4)^{1/30} = 1.0296$

$\therefore r = 1.0296 - 1 = 0.0296$

$= 2.96\%$

- **Example 4:** Suppose one has an investment that doubles in value in the first year, and then falls back to the original value.
- What would be the average rate of return?
- Doubling from X to $2X$ is a rate of return of $+100\%$.
- Falling back to X from $2X$ is a rate of return of -50%.
- Average rate of return is 25%.

First year $\frac{200-100}{100} = 1$ i.e. 100%

Second year $\frac{100-200}{200} = \frac{-1}{2} = -50\%$.

$\therefore$ Average $= \frac{100 - 50}{2} = 25\%$.

- **Example 5:** Consider an investment that gives a rate of return of $+20\%$ followed by a rate of return of -20%.
- This would have an average rate of return of 0%.
- However, in reality it would leave you will a loss.
- An investment of 1000 would become 1200 after first year, but drop down to 960 in the second year.
- The average rate of return of 0% does not accurately highlight this loss.
- If you calculate the annualized rate $r_2 = \sqrt{1-4\%} - 1 \approx -2.02\%$, it would show that one had a loss.

First year $\frac{120-100}{100} = 0.2$ i.e. 20%.

Second year loss of 20% i.e. $120 - \frac{2}{10} \times 120 = 96$.

$\therefore$ After 2 years = 96.

$\therefore \frac{96-100}{100} = \frac{-4}{100} = -0.04$.

$\therefore 1 - 0.04 = (1+r)^2$

$\therefore 1+r = (0.96)^{1/2} =$

$\therefore r = -0.02$ ✓

Imp

- With time varying interest rates, the NPV formula becomes:

$$\begin{aligned} NPV &= PV(C_0) + PV(C_1) + PV(C_2) + \dots \\ &= C_0 + \frac{C_1}{1+r_{0,1}} + \frac{C_2}{1+r_{0,2}} + \dots \\ &= C_0 + \frac{C_1}{1+r_{0,1}} + \frac{C_2}{(1+r_{0,1})(1+r_{1,2})} + \dots \\ &= C_0 + \frac{C_1}{1+r_1} + \frac{C_2}{(1+r_2)^2} + \dots \end{aligned}$$

- **Example 6:** Consider a project with an initial investment of 24 that pays 20 in one year and 16 in five years.
- Assume that the 1-year interest rate is 5% and the five-year annualized interest rate is 6% per annum.
- PV of 20 in one year is equal to $20/(1.05)$.
- PV of 16 in five years is equal to $16/(1.06)^5$.
- NPV of the project is equal to

$$\begin{aligned} NPV &= -24 + \frac{20}{1.05} + \frac{16}{(1.06)^5} \\ &\approx 7.01. \end{aligned}$$

$NPV = -24 + \frac{20}{1+0.05} + \frac{16}{(1.06)^5} \approx 7.01$

GAs

1) Suppose SBI offers a home loan of Rs. 800,000 at 9% annual interest rate. A borrower takes the loan and agrees to repay in 20 years with equal monthly payments (beginning from next month). How much does the borrower need to pay each month? (use banking convention of multiplying monthly interest rate by 12 to get annual rate) (Rounded off to the nearest integer)

- ○ Rs 6,000
- ○ Rs 7,303
- ○ Rs 6,849
- ✓ Rs 7,198

Monthly interest = $\frac{0.09}{12} \times 100 = \frac{9}{12} = 0.75\%$ ✓

$800000 = \frac{C}{0.0075}\left(1 - \frac{1}{1.0075^{240}}\right)$

$\therefore C = \frac{800000 \times 0.0075 \times 1.0075^{240}}{1.0075^{240} - 1} = 7197.8 \approx 7198$

2) Suppose SBI offers a home loan of Rs. 800,000 at 9% annual interest rate. A borrower takes the loan and agrees to repay in 20 years with **5 poi** equal monthly payments (beginning from next month). The borrower wins a lottery after 10 years (after 120 months) and wants to pay off the loan immediately. How much would he need to pay at that time? (use banking convention of multiplying monthly interest rate by 12 to get annual rate) (Rounded off to the nearest integer)

- ○ 794,613
- ✓ 568,170
- ○ 622,523
- ○ 469,781

monthly payment = 7198.

how much he has paid in 120 months

$$= \frac{7198}{0.0075}\left(1 - \frac{1}{1.0075^{120}}\right) = 568222$$

The formula for the remaining loan balance after n payments is:

Imp. formula

$$B_n = P \times \frac{(1+i)^N - (1+i)^n}{(1+i)^N - 1}$$

$$= 800000 \times \frac{(1.0075)^{240} - (1.0075)^{120}}{(1.0075)^{240} - 1}$$

$$= 568207 \checkmark$$

3) ABC Limited has a forward-looking dividend yield of 3.5%. ABC Limited pays a dividend of Rs 50 per share this year. The dividend is expected to grow by 5% each year. What is the estimated appropriate cost of capital for ABC Limited?

- ○ 1.5%
- ○ 3.5%
- ✓ 8.5%
- ○ 12.5%

forward-looking dividend yield = $r - g$.

$\therefore 3.5 = r - 5$

$\therefore r = 5 + 3.5 = 8.5$

4) Ram and Shyam are two friends, and both have income of 250 units per period. Ram is a person who tries to save as much as possible for **5 poir** the future while Shyam believes in consuming maximum in the current period. An investment opportunity appears which demands 250 units in the current period and returns 300 units in the next period. If the constant interest rate is 15% per period, then

- ✓ Ram and Shyam both should take the investment opportunity.
- ○ Ram should take the investment opportunity while Shyam should avoid the investment opportunity.
- ○ Ram should avoid the investment opportunity while Shyam should take the investment opportunity.
- ○ Ram and Shyam both should avoid the investment opportunity.

Ram if he invests 250 he gets 300 in next period.

he puts 250 in bank, he gets $250 + \frac{15}{100} \times 250 = 250 + 37.5 = 287.5$

$\therefore$ he should invest.

Shyam purchases for 250 & sells at present value $= \frac{30000}{115} = 260.86 > 250$

So he should also invest.

$\therefore$ Both should invest.

5) You invest in a project which pays Rs 500 next year and Rs 500 two years from today. The correct constant interest rate is 10% per year. But you make an error in your estimation and assume that project will pay Rs 600 in both periods, then **5**

- [x] You made a 20% error in cash flow calculation which resulted in you calculating 20% higher present value of the project.
- [ ] You made a 20% error in cash flow calculation which resulted in you calculating 20% lower present value of the project.
- [ ] You made a 20% error in cash flow calculation which resulted in you calculating 10% higher present value of the project.
- [ ] You made a 10% error in cash flow calculation which resulted in you calculating 10% higher present value of the project.

$NPV_1 = \frac{500}{1.1} + \frac{500}{1.1^2} = 867.76.$

$NPV_2 = \frac{600}{1.1} + \frac{600}{1.1^2} = 1041.32 = 867.76 + 173.56$

$= 867.76 + 0.2 \times 867.76$

error in Cash flow : $\frac{600-500}{500} : \frac{100}{500} \times 100$ 20% ✓ ∴ 20% more.

6) What is the IRR of a project that costs Rs 50,000 today and pays Rs 60,000 four years from today.?

- [ ] 20.00%
- [ ] 5.00%
- [x] 4.66%
- [ ] 7.18%

$-50000 + \frac{60000}{(1+r)^4} = 0$, let $1+r = x$.

$\therefore +50000 x^4 = +60000 \quad \therefore x^4 = \frac{6}{5} \quad \therefore x = \left(\frac{6}{5}\right)^{1/4}$

$\therefore r = \left(\frac{6}{5}\right)^{1/4} - 1 = 1.0466 - 1 = 0.0466 = 4.66\%.$

7) What is the yield-to-maturity (YTM) of a two-year coupon bond that costs Rs 10,000 today and pays Rs 2,000 at the end of each year in coupon payments and also pays the value of Rs 8,000 at the end of the second year.

- [x] 10.5%
- [ ] 12.5%
- [ ] 15.0%
- [ ] 20.0%

IRR is also known as 'yield to majority'.

$-10000 + \frac{2000}{1+x} + \frac{2000}{(1+x)^2} + \frac{8000}{(1+x)^2} = 0$

$\Rightarrow 10(1+x)^2 - 2(1+x) - 10 = 0 \Rightarrow 10(1+x^2+2x) - 2 - 2x - 10 = 0$

$\Rightarrow 10 + 10x^2 + 20x - 2 - 2x - 10 = 0$

$\Rightarrow 10x^2 + 18x - 2 = 0 \quad \therefore x = \frac{-18 \pm \sqrt{18^2 + 80}}{20}$

$\therefore x = 0.1049 = 10.49\%$

8) Cobra Inc. is considering two different projects A and B for investment. Project A costs Rs 95,000 and is expected to generate Rs 65,000 in year one and Rs 75,000 is year two. Project B costs Rs 120,000 and is expected to generate Rs 64,000 in year one, Rs 67,000 in year two, Rs 56,000 in year three, and Rs 45,000 in year four. Cobra Inc.'s required rate of return for these projects is 10%. The net present value for Project A is (Rounded off to the nearest integer) ***5 points***

- [ ] 16,947
- [ ] 19,458
- [ ] 22,845
- [x] 26,074

$NPV_1 = -95000 + \frac{65000}{1.1} + \frac{75000}{(1.1)^2} = 26074$

9) Cobra Inc. is considering two different projects A and B for investment. Project A costs Rs 95,000 and is expected to generate Rs 65,000 in year one and Rs 75,000 is year two. Project B costs Rs 120,000 and is expected to generate Rs 64,000 in year one, Rs 67,000 in year two, Rs 56,000 in year three, and Rs 45,000 in year four. Cobra Inc.'s required rate of return for these projects is 10%. The net present value for Project B is (Rounded off to the nearest integer) **5 points**

- 58,097
- ✓ 66,363
- 74,538
- 86,465

$NPV_2 = -120000 + \frac{64000}{1.1} + \frac{67000}{1.1^2} + \frac{56000}{1.1^3} + \frac{45000}{1.1^4}$

$= 66362.8$

10) Cobra Inc. is considering two different projects A and B for investment. Project A costs Rs 95,000 and is expected to generate Rs 65,000 in year one and Rs 75,000 is year two. Project B costs Rs 120,000 and is expected to generate Rs 64,000 in year one, Rs 67,000 in year two, Rs 56,000 in year three, and Rs 45,000 in year four. Cobra Inc.'s required rate of return for these projects is 10%. The profitability index for Project A and Project B is respectively **5 points**

- ✓ 1.27 and 1.55
- 1.17 and 1.39
- 1.33 and 1.65
- 1.08 and 1.25

$PI_1 = \frac{26074 + 95000}{95000} = 1.27$

$PI_2 = \frac{66362 + 120000}{120000} = 1.55$

11) Cobra Inc. is considering two different projects A and B for investment. Project A costs Rs 95,000 and is expected to generate Rs 65,000 in year one and Rs 75,000 is year two. Project B costs Rs 120,000 and is expected to generate Rs 64,000 in year one, Rs 67,000 in year two, Rs 56,000 in year three, and Rs 45,000 in year four. Cobra Inc.'s required rate of return for these projects is 10%. The internal rate of return for Project A is **5 points**

- 31.73%
- ✓ 29.42%
- 25.88%
- 19.45%

$-95000 + \frac{65000}{1+r} + \frac{75000}{(1+r)^2} = 0 \Rightarrow -19 + \frac{13}{1+r} + \frac{15}{(1+r)^2} = 0$

$19(1 + r^2 + 2r) - 13 - 13r - 15 = 0$

$19r^2 + 38r - 13r + 19 - 13 - 15 = 0$

$\therefore 19r^2 + 25r - 9 = 0 \quad \therefore r = (-25 \pm \sqrt{625 + 36 \times 19})/38$

$= 0.294 = 29.4\%$

12) Cobra Inc. is considering two different projects A and B for investment. Project A costs Rs 95,000 and is expected to generate Rs 65,000 in year one and Rs 75,000 is year two. Project B costs Rs 120,000 and is expected to generate Rs 64,000 in year one, Rs 67,000 in year two, Rs 56,000 in year three, and Rs 45,000 in year four. Cobra Inc.'s required rate of return for these projects is 10%. The internal rate of return for Project B is **5 points**

- 29.74%
- 30.79%
- ✓ 35.27%
- 38.68%

$-120000 + \frac{64000}{1+r} + \frac{67000}{(1+r)^2} + \frac{56000}{(1+r)^3} + \frac{45000}{(1+r)^4} = 0$

$-120(1+r)^4 + 64(1+r)^3 + 67(1+r)^2 + 56(1+r) + 45 = 0$

use Excel to solve. $= 0.35 = 35\%$.

13) Vyom Corporation is considering a new system that will cost Rs 750,000. The system is expected to generate positive cashflows over t next four years in the amounts of Rs 350,000 in year one, Rs 325,000 in year two, Rs 150,000 in year three, and Rs 180,000 in year four. Ass the payment is equally distributed throughout a year. The company's required rate of return is 8%. What is the payback period of this systen

- 4.00 years
- 3.09 years
- 2.91 years
- ✓ 2.50 years

$\text{payback} = 1 + 1 + 0.5 = 2.5$

14) You bought a painting 12 years ago for Rs 30,000. Today you sell it for Rs 100,000. What is the annualized rate of return that you earned on the painting?

- 8.64%
- ✓ 10.55%
- 12.84%
- 27.78%

Total return $= \frac{100000 - 30000}{30000} = \frac{70000}{30000} = \frac{7}{3} = 2.3333$

$\therefore \quad 1 + r_t = (1+r)^{12} = 3.33 \quad \therefore \quad 1 + r = 3.33^{1/12}$

$\therefore r = 3.33^{1/12} - 1 = 10.54\%$

15) A project earns a total of 40% return in three years. If annual return in first years is 15%, and annual return in the second year is 12%. then what is the annual rate of return in the third year?

- 13.00%
- 11.42%
- ✓ 8.70%
- 6.36%

$1.4 = (1.15)(1.12)(1+r_3)$

$\therefore 1 + r_3 = \frac{1.4}{1.15 \times 1.12} \quad \therefore r_3 = \frac{1.4}{1.15 \times 1.12} - 1 = 0.869$

$= 8.69\%$

16) Suppose the share price of XYZ Corp. is Rs 150 today. The XYZ share price rises by 30% in the next period and then falls by 20% in the second period. What is the share price of XYZ Corp. after two periods?

- 195
- 165
- ✓ 156
- 150

After 1 year $= 150 + 0.3 \times 150 = 150 + 45 = 195$

after 2 per $= 195 - 0.2 \times 195 = 195 - 39 = 156$

17) Consider the following cashflows for projects and rates of return for different periods.

	Period 1	Period 2	Period 3	Period 4
Rate of return	**8%**	**12%**	**10%**	**8%**
Project 1	1000	1200	1300	1500
Project 2	800	1200	1400	1600
Project 3	0	2500	0	2500
Project 4	0	0	0	5000

$NPV_1 = -3500 + \frac{1000}{1.08} + \frac{1200}{1.08 \times 1.12} + \frac{1300}{1.08 \times 1.12 \times 1.1} + \frac{1500}{1.08 \times 1.12 \times 1.1 \times 1.08} = 438.$

$NPV_2 = -3500 + \frac{800}{1.08} + \frac{1200}{1.08 \times 1.12} + \frac{1400}{1.08 \times 1.12 \times 1.1} + \frac{1600}{1.08^2 \times 1.12 \times 1.1} = 398.$

$NPV_3 = -3500 + \frac{2500}{1.08 \times 1.12} + \frac{2500}{1.08^2 \times 1.1 \times 1.12} = 306$

$NPV_4 = -3500 + \frac{5000}{1.08^2 \times 1.1 \times 1.12} = -20.$

If each project requires an initial investment of Rs 3500 in the current period, and you can choose only one project, then which of the project should be chosen according to the NPV capital budgeting rule?

- ✓ Project 1
- Project 2
- Project 3
- Project 4

If each project requires an initial investment of Rs 3500 in the current period, then which of the project has the lowest internal rate of return?

- Project 1 → $-3500 + \frac{1000}{1+r} + \frac{1200}{(1+r)^2} + \frac{1300}{(1+r)^3} + \frac{1500}{(1+r)^4} = 0$
- Project 2 →
- Project 3
- ✓ Project 4

Using Excel IRR = 15, 14, 13, 9

$\therefore$ Project 4 has lowest.

If each project requires an initial investment of Rs 3500 in the current period, then which of the following statement(s) is/are correct?

- ☐ All projects have a positive net present value (NPV).
- ✓ ☑ All projects have a positive internal rate of return (IRR).
- ✓ ☑ Some projects have a negative net present value (NPV).
- ✓ ☑ The preference order is same under the NPV capital budgeting rule and the IRR capital budgeting rule.

Yes. 1234

20) Infinity Inc. has 100,000 shares outstanding which are trading at Rs 100 per share. Infinity Inc. is expected to earn Rs 500,000 as profit next year. The profit is expected to grow at 2% per annum thereafter. The firm faces a cost of capital of 8% per annum. If the firm pays a constant dividend of Rs 15 per year per share beginning from next year, then choose the correct statement(s) about Infinity Inc. **5 points**

- [x] According to the growing perpetuity formula on profit, the shares of Infinity Inc. are overpriced.
- [ ] According to the growing perpetuity formula on profit, the shares of Infinity Inc. are underpriced.
- [ ] According to the Gordon growth model, if the appropriate rate of return is 12%, then the shares of Infinity Inc. are overpriced.
- [x] According to the Gordon growth model, if the appropriate rate of return is 12%, then the shares of Infinity Inc. are underpriced.

As per growing perpetuity., Valuation $= \frac{C}{r-g} = \frac{500000}{8-2} = \frac{50000000}{0.06}$

$\therefore$ every share value will be $\frac{50000000}{6 \times 100000} = 83.33$

$\therefore$ At 100 Rs., overpriced.

Gordon growth model $\frac{d}{P} = r - g$, $g = 8\%$, $r = 12\%$.

$\therefore P = \frac{d}{0.04} = \frac{1500}{0.04} = 375$

$\therefore$ at 100 Rs, underpriced.